

# Chat History: CompTIA CySA+ CS0-003 Study Notes and Foundation Guide

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**Exported:** 2026-07-02T16:54:08.231155

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**You (Tue Jun 30 17:50:24 2026)**

just confirm is there tcp dump commands taught here ....just answer yes or np

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**NotebookLM (Tue Jun 30 17:50:24 2026)**

**No.**

While **tcpdump** is defined as a command-line packet analyzer [1] and mentioned as a log output format for other tools [2], specific **tcpdump commands** (syntax or usage examples) are not taught in these sources. In contrast, the notes do provide specific command syntax for tools like **Nmap** [3, 4], **hping3** [5, 6], and **grep** [7].

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** "tcpdump\*\* is defined as a command-line packe" - [2] **Source 2:** " analyzer [1] and mentioned as a log output format f" - [3] **Source 3, Source 4** - [4] **Source 5, Source 6** - [5] **Source 7**

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**You (Tue Jun 30 17:51:45 2026)**

Question 5 Incorrect Which of the following is the correct usage of the tcpdump command to create a packet capture filter for all traffic going to and from the server located at 10.10.1.1?

any relevant infirmation given i source for solving this question

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## NotebookLM (Tue Jun 30 17:51:45 2026)

The provided sources define **tcpdump** and its purpose but do **not** contain specific command-line syntax or examples for its usage.

According to the sources, the relevant information regarding **tcpdump** is as follows:

\* **Definition:** It is a data-network packet analyzer computer program that runs under a command-line interface [1]. \* **Purpose:** It allows a user to display TCP/IP and other packets being transmitted or received over a network to which the computer is attached [1]. \* **Context:** It is mentioned as a log output format for other tools like Snort [2] and as a potential data source for Yara rules [3].

**Information from outside the sources:** Since the syntax is not in your study notes, you may want to independently verify this, but the correct `tcpdump` usage to capture all traffic going to and from a specific IP address is:  
`tcpdump host 10.10.1.1`

Using the `host` keyword acts as a bidirectional filter for both the source and destination of the packets. If you only wanted traffic *from* the server, you would use `src`, and if you only wanted traffic *to* the server, you would use `dst`.

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** “ws: \* **Definition:** It is a data-network packet analyzer computer program that runs under” - [2] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** “line interface [1]. \* **Purpose:** It allows a user to display TCP/IP and other packets being tran...” - [3] **Source 2:** “ch the computer is attached [1]. \* **Context:** It is mentioned” - [4] **Source 3:** “as a log output format for other tools like S”

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## You (Tue Jun 30 17:54:31 2026)

tell me there is a table related to encoding in the soources if so please draw that table here

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## NotebookLM (Tue Jun 30 17:54:31 2026)

Yes, the sources contain a table specifically related to **percent encoding** (also known as URL encoding), which is used to encode 8-bit characters that have specific meanings in the context of URLs [1, 2].

Here is the table as it appears in the sources:

| Character | Percent Encoding |
|-----------|------------------|
| null      | %00              |
| space     | %20              |
| +         | %2B              |
| %         | %25              |
| /         | %2F              |
| \         | %5C              |
| .         | %2E              |
| ?         | %3F              |
| "         | %22              |
| '         | %27              |
| <         | %3C              |
| >         | %3E              |

[2]

In addition to this table, the sources also mention other forms of encoding, such as:

- \* **Base64 encoding:** Used for Windows portable executable file headers (MZ in ASCII is TV in Base64) and for encoding binary data within email bodies using MIME [3-5].
- \* **Unicode encoding:** Mentioned as a format that the Strings tool can extract from executable files [6].
- \* **UTF-8/ASCII:** The sources note that URLs can generally only contain unreserved (a-z, A-Z, 0-9, etc.) and reserved characters from the ASCII set [1].

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**, **Source 2:** "g\*\* (also known as URL encoding), which is used to encode 8-bit characters that have specific meanin..." - [2] **Source 2** - [3] **Source 3, Source 4, Source 5:** " | %27 | | < | %3C | | > | %3E |

[2]

In addition to this table, the sources also mention other for..." - [4] **Source 6:** " Windows portable executable file headers (MZ in ASCII is TV in Base64) and f" - [5] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** "nary data within email bodies using MIME [3-5]. \* **Unicode encoding:** Mentioned as a format that..."

## You (Tue Jun 30 17:57:29 2026)

id=1%20OR%2017-7%3d10.

could u tell me whats this %3d here

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## NotebookLM (Tue Jun 30 17:57:29 2026)

In the string you provided, `%3d` is the **percent-encoded** (or URL-encoded) representation of the **equals sign (=)**. [1, 2]

While the specific code for the equals sign is not listed in the summary table within the sources, the sources define **percent encoding** as a mechanism to encode characters that have specific meanings in the context of URLs. [1] The sources explicitly list the equals sign (=) as a **reserved character** [1], and following the pattern of other encoded characters in the sources (such as `<` being `%3C` and `>` being `%3E`), `%3D` is the standard hex representation for `=`. [2]

When decoded, your string `id=1%20OR%2017-7%3d10` translates to: `id=1 OR 17-7=10`

Here is a breakdown of the encoding used in that string based on the sources: \* `%20` : Represents a **space**. [2] \* `%3d` : Represents the **equals sign (=)**. [1, 2]

## Contextual Insight

This specific string is an example of an **SQL injection** attempt. [3, 4] Attackers use encoded strings like this in URL parameters to bypass security. The statement `17-7=10` is a **tautology** (a statement that is always true), which is a common technique used in injection attacks to manipulate database queries into returning unauthorized data or granting access. [3, 4]

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**, **Source 2:** "a" - [2] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**: "efine **percent encoding** as a mechanism to encode characters that have specific meani" - [3] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf** - [4] **Source 2:** "E" - [5] **Source 2:** "d" - [6] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**, **Source 2:** "e" - [7] **Source 3**, **Source 4:** "sents th" - [8] **Source 3**, **Source 4:** "empt. [3, 4] Attackers use encoded strings like this in URL parameters to bypass security. The state..."

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## You (Tue Jun 30 17:59:07 2026)

is there some more common encoding which i must know in order to solve such attacks from happening by detecting them...if so just give me a simple table

## NotebookLM (Tue Jun 30 17:59:07 2026)

Based on the sources, while **percent encoding** is the most frequently detailed due to its use in web-based attacks, there are several other common encoding and representation formats you must recognize to detect obfuscation and injection attempts.

Below is a table of the most common encoding types and their contexts as mentioned in the sources:

| Encoding Type            | Description / Context                                                                                               | Detection / Usage Insight                                                                                                                                                                                  |
|--------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Percent (URL)</b>     | Mechanism to encode characters with specific meanings in URLs (e.g., <code>.</code> is <code>%2e</code> ). [1], [2] | Used to obfuscate <b>directory traversal</b> (e.g., <code>%2e%2e%2f</code> for <code>../</code> ) or to hide malicious scripts. [3], [4]                                                                   |
| <b>Base64</b>            | Encodes binary data into a sequence of printable ASCII characters. [5]                                              | Used in <b>MIME email bodies</b> to send attachments or binary data. [6], [5] Also used in file headers (e.g., a Windows executable's <code>MZ</code> signature appears as <code>TV</code> in Base64). [7] |
| <b>Hexadecimal (Hex)</b> | A base-16 representation of data (0-9, A-F). Machine code is typically represented as two hex digits per byte. [7]  | Used to <b>obfuscate data</b> in covert channels or represent file signatures (e.g., Windows PE files start with <code>4D 5A</code> ). [7], [8], [9]                                                       |
| <b>Unicode</b>           | A standard for encoding characters from virtually all languages. [10]                                               | Analysts use the <b>Strings tool</b> to extract Unicode sequences (over three characters long) from executable files to find hidden functionality. [10]                                                    |

## Advanced Detection Considerations

Beyond identifying these formats, you should be aware of specific attack techniques that leverage them:

- **Double-Encoding:** Tricky attackers may encode the percent sign itself (e.g., encoding `%` as `%25` ) to bypass simple security filters that only look for a single layer of encoding. [3]
- **Canonicalization Attacks:** These occur when an attacker uses encoding (like hex or percent encoding) to submit input that evades a system's input validation measures. [11]
- **Malware Obfuscation:** Malicious code often uses compression, encryption, or **encoding** specifically to hide its logic from static analysis and signature-matching tools. [12], [13], [14]

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** "te" - [2] **Source 2:** "r" - [3] **Source 3:** "]" | Used to obfuscate **directo**" - [4] **Source 4:** "r" - [5] **Source 5:** "ersal (e.g., `%2e%2e%2f` for `../` ) or to hide malicious scripts" - [6] **Source 6:** "]" Encodes binary data into a sequen" - [7] **Source 5:** "c" - [8] **Source 7:** "ents or bin" - [9] **Source 7:** "]" Also used in file headers (e.g., a Windows executable's `MZ` signature appears as `TV` in Base64)...." - [10] **Source 7:** "o" - [11] **Source 8:** "h" - [12] **Source 9:** "e" - [13] **Source 10:** "s per byte. [7] | Used to **obfuscate data** in covert channels" - [14] **Source 10:** "e.g., Windows PE files start with `4D 5A` ). [7], [8], [9] | | **Unicode** | A standard for encoding ..." - [15] **Source 3:** "Beyond identifying these formats, you should be aware of specific attack technique" - [16] **Source 11:** " **Double-Encoding:** Tricky attackers may encode the percent sign itself (e.g., encoding `%` as ``%2...`" - [17] **Source 12:** "These occur when an attacker uses encoding (like hex or percent encoding) to subm" - [18] **Source 13:** "i" - [19] **Source 14:** "t"

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## You (Wed Jul 1 14:06:23 2026)

i want u to help me with the understanding of below questions  
grep-F 192.168.1.10 access.log  
grep ● "192.168.1.10" \*  
grep-r ● 192.168.1.{1,3} .  
grep-r 192.168.1.[0-255]

and what is this `0?[0-9]{1,2}` while using regex to search for an ip address please give me simple and brief explanation if all

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## NotebookLM (Wed Jul 1 14:06:23 2026)

Based on the provided study notes, here is a simple and brief explanation of the `grep` commands and the regex pattern you mentioned.

### Understanding the `grep` Commands

The `grep` command is used on Unix/Linux systems to search text files for specific strings or patterns [1].

- `grep -F 192.168.1.10 access.log` : This searches the file named `access.log` for the **literal string** “192.168.1.10” [1]. The `-F` option (Fixed strings) tells `grep` to treat the pattern as a simple string rather than a regular expression.
- `grep "192.168.1.10" *` : This searches for the string “192.168.1.10” in **every file** ( `*` ) within the current directory [1].
- `grep -r 192.168.1.[\d]{1,3} .` : This performs a **recursive search** ( `-r` ) starting from the current directory ( `.` ) and looking into all subdirectories [1]. It uses regex to find any IP address starting with `192.168.1.` followed by **1 to 3 digits** ( `[\d]{1,3}` ) [2, 3].
- `grep -r 192.168.1.[1-255] .` : This is also a recursive search, but it contains a common regex error. The brackets `[...]` match a **single instance** of a character [2]. Therefore, `[1-255]` will only match a **single digit** that is either 0, 1, 2, or 5; it does not match the numerical range 0 through 255 [1, 2].

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### Regex Explanation: `0?[4-12]{1,2}`

When searching for an IP address, this pattern breaks down as follows:

- `0?` : The `?` quantifier means “one or none” [3]. This makes a leading zero **optional** (it matches “0” or nothing).
- `[4-12]` : The brackets match a **single digit** between 0 and 9 [2].
- `{1,2}` : The curly braces are a quantifier meaning the previous element (the digit) must occur **one or two times** [3].

**In simple terms:** This pattern matches a number that is **one or two digits long** (like 5, 55, or 99) and allows it to optionally start with a zero (like 05) [2, 3]. Analysts use these quantifiers to ensure they capture different variations of octets within an IP address.

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**: “s The `grep` command is used on Unix/Linux systems to search text files for specific

string" - [2] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**: "ss.log for  
th" - [3] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**: "earches for the  
string "192.16" - [4] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**: "his  
performs a **recursive search**" - [5] **Source 2, Source 3**: "e" - [6]  
**CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**, Source 2, Source 3, Source 4,  
Source 5, Source 6, Source 7, Source 8, Source 9, Source 10, Source 11, Source  
12, Source 13, Source 14, Source 15, Source 16, Source 17, Source 18, Source  
19, Source 20, Source 21, Source 22, Source 23, Source 24, Source 25, Source  
26, Source 27, Source 28, Source 29, Source 30, Source 31, Source 32, Source  
33, Source 34, Source 35, Source 36, Source 37, Source 38, Source 39, Source  
40, Source 41, Source 42, Source 43, Source 44, Source 45, Source 46, Source  
47, Source 48, Source 49, Source 50, Source 51, Source 52, Source 53, Source  
54, Source 55, Source 56, Source 57, Source 58, Source 59, Source 60, Source  
61, Source 62, Source 63, Source 64, Source 65, Source 66, Source 67, Source  
68, Source 69, Source 70, Source 71, Source 72, Source 73, Source 74, Source  
75, Source 76, Source 77, Source 78, Source 79, Source 80, Source 81, Source  
82, Source 83, Source 84, Source 85, Source 86, Source 87, Source 88, Source  
89, Source 90, Source 91, Source 92, Source 93, Source 94, Source 95, Source  
96, Source 97, Source 98, Source 99, Source 100, Source 101, Source 102,  
Source 103, Source 104, Source 105, Source 106, Source 107, Source 108, Source  
109, Source 110, Source 111, Source 112, Source 113, Source 114, Source 115,  
Source 116, Source 117, Source 118, Source 119, Source 120, Source 121, Source  
122, Source 123, Source 124, Source 125, Source 126, Source 127, Source 128,  
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135, Source 136, Source 137, Source 138, Source 139, Source 140, Source 141,  
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Source 207, Source 208, Source 209, Source 210, Source 211, Source 212, Source  
213, Source 214, Source 215, Source 216, Source 217, Source 218, Source 219,  
Source 220, Source 221, Source 222, Source 223, Source 224, Source 225, Source  
226, Source 227, Source 228, Source 229, Source 230, Source 231, Source 232,  
Source 233, Source 234, Source 235, Source 236, Source 237, Source 238, Source  
239, Source 240, Source 241, Source 242, Source 243, Source 244, Source 245,



Source 246, Source 247, Source 248, Source 249, Source 250, Source 251, Source 252, Source 253, Source 254, Source 255\*\*: " to find any IP ad" - [7] \*\*Source 2\*\*: ". : **This is** " - **[8]** CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf, Source 2, Source 3, Source 4, Source 5, Source 6, Source 7, Source 8, Source 9, Source 10, Source 11, Source 12, Source 13, Source 14, Source 15, Source 16, Source 17, Source 18, Source 19, Source 20, Source 21, Source 22, Source 23, Source 24, Source 25, Source 26, Source 27, Source 28, Source 29, Source 30, Source 31, Source 32, Source 33, Source 34, Source 35, Source 36, Source 37, Source 38, Source 39, Source 40, Source 41, Source 42, Source 43, Source 44, Source 45, Source 46, Source 47, Source 48, Source 49, Source 50, Source 51, Source 52, Source 53, Source 54, Source 55, Source 56, Source 57, Source 58, Source 59, Source 60, Source 61, Source 62, Source 63, Source 64, Source 65, Source 66, Source 67, Source 68, Source 69, Source 70, Source 71, Source 72, Source 73, Source 74, Source 75, Source 76, Source 77, Source 78, Source 79, Source 80, Source 81, Source 82, Source 83, Source 84, Source 85, Source 86, Source 87, Source 88, Source 89, Source 90, Source 91, Source 92, Source 93, Source 94, Source 95, Source 96, Source 97, Source 98, Source 99, Source 100, Source 101, Source 102, Source 103, Source 104, Source 105, Source 106, Source 107, Source 108, Source 109, Source 110, Source 111, Source 112, Source 113, Source 114, Source 115, Source 116, Source 117, Source 118, Source 119, Source 120, Source 121, Source 122, Source 123, Source 124, Source 125, Source 126, Source 127, Source 128, Source 129, Source 130, Source 131, Source 132, Source 133, Source 134, Source 135, Source 136, Source 137, Source 138, Source 139, Source 140, Source 141, Source 142, Source 143, Source 144, Source 145, Source 146, Source 147, Source 148, Source 149, Source 150, Source 151, Source 152, Source 153, Source 154, Source 155, Source 156, Source 157, Source 158, Source 159, Source 160, Source 161, Source 162, Source 163, Source 164, Source 165, Source 166, Source 167, Source 168, Source 169, Source 170, Source 171, Source 172, Source 173, Source 174, Source 175, Source 176, Source 177, Source 178, Source 179, Source 180, Source 181, Source 182, Source 183, Source 184, Source 185, Source 186, Source 187, Source 188, Source 189, Source 190, Source 191, Source 192, Source 193, Source 194, Source 195, Source 196, Source 197, Source 198, Source 199, Source 200, Source 201, Source 202, Source 203, Source 204, Source 205, Source 206, Source 207, Source 208, Source 209, Source 210, Source 211, Source 212, Source 213, Source 214, Source 215, Source 216, Source 217, Source 218, Source 219, Source 220, Source 221, Source 222, Source 223, Source 224, Source 225, Source 226, Source 227, Source 228, Source 229, Source 230, Source 231, Source 232, Source 233, Source 234, Source 235, Source 236, Source 237, Source 238, Source 239, Source 240, Source 241, Source 242, Source 243, Source 244, Source 245, Source 246, Source 247, Source 248, Source 249, Source 250, Source 251, Source 252, Source 253, Source 254, Source 255 - **[9]**

CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf, Source 2: "mmon regex error. The brackets [...] match a single instance\*\* of a character" - [10] Source 4, Source 5, Source 6, Source 7, Source 8, Source 9, Source 10, Source 11, Source 12: "25" - [11] Source 3: "h the numerical range 0 through" - [12] Source 4, Source 5, Source 6, Source 7, Source 8, Source 9, Source 10, Source 11, Source 12 - [13] Source 2: "tern breaks dow" - [14] Source 3 - [15] Source 2, Source 3: "digit\*\* between 0 and 9 [2]. \* {1,2} : The curly braces are a quanti"

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## You (Wed Jul 1 14:10:02 2026)

0?[0-9]{1,2} together what is this doing like [0-9] what is this doing

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## NotebookLM (Wed Jul 1 14:10:02 2026)

Based on the regex elements described in the sources, here is a breakdown of what 0?[1-9]{1,2} is doing:

### 1. [1-9] (The Character Class)

The brackets [...] match a **single instance** of any character listed inside them [10]. Therefore, [1-9] matches any **single digit** between 0 and 9 [10].

### 2. 0? (The Optional Zero)

The ? is a quantifier that means "one or none" [11]. Placing it after a 0 makes that zero **optional**. It will match "0" if it's there, but it will also match if no zero is present [11].

### 3. {1,2} (The Range Quantifier)

The curly braces { } specify how many times the preceding element must occur [11]. In this case, {1,2} means the preceding digit ( [1-9] ) must appear **at least one time and at most two times** [11].

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## Summary: What the whole pattern does together

When you put it all together, `0?[1-9]{1,2}` searches for a number that is **one or two digits long** and allows for a **leading zero**.

- **It will match:** "5", "55", "05", or "055".
- **It will NOT match:** "555" (too many digits for the `{1,2}` part) or "A" (not a digit).

In cybersecurity, analysts use patterns like this to find IP address octets in log files, accounting for the fact that some logs might represent a number like "5" as "05" or "005" [12].

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf, Source 2, Source 3, Source 4, Source 5, Source 6, Source 7, Source 8, Source 9:** "" - [2] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf, Source 2, Source 3, Source 4, Source 5, Source 6, Source 7, Source 8, Source 9** - [3] **Source 10\*\*:** "" **[...] match a single instanc**" - [4] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf, Source 2, Source 3, Source 4, Source 5, Source 6, Source 7, Source 8, Source 9** - [5] **Source 10:** " [10]. Therefor" - [6] **Source 11:** "gle digit\*\* between 0 and 9 [10].

**\*"**

- [7] **Source 11:** " quantifier that means "one or none" [11]. Placing it after a `0` makes that ze"
- [8] **Source 11:** " it will also match if no zero is present [11].

**\*\*"**

- [9] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf, Source 2, Source 3, Source 4, Source 5, Source 6, Source 7, Source 8, Source 9**
- [10] **Source 11**
- [11] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf, Source 2, Source 3, Source 4, Source 5, Source 6, Source 7, Source 8, Source 9:** "[1"
- [12] **Source 12:** "or two digits long\*\* and allows for a **leading zero**.
- **It will match:** "5", "55", "05", or ..."

**You (Thu Jul 2 12:46:45 2026)**

in the sources there where names of some ports with refercnce to the protocols assigned could u please make me a table for them

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**NotebookLM (Thu Jul 2 12:46:45 2026)**

Based on the provided source material, here is a consolidated table of the **TCP and UDP ports** that a cybersecurity analyst should know, as well as the protocols assigned to them.

## **Commonly Scanned Network Ports**

| Port       | Transport Protocol | Service Name      | Description                                                         |
|------------|--------------------|-------------------|---------------------------------------------------------------------|
| <b>21</b>  | TCP                | <b>FTP</b>        | File Transfer Protocol [1]                                          |
| <b>22</b>  | TCP                | <b>SSH/SFTP</b>   | Secure Shell / FTP over SSH [1]                                     |
| <b>23</b>  | TCP                | <b>TELNET</b>     | Unsecure remote administration interface [1]                        |
| <b>25</b>  | TCP                | <b>SMTP</b>       | Simple Mail Transfer Protocol [1]                                   |
| <b>53</b>  | TCP / UDP          | <b>DNS</b>        | Domain Name System (TCP for zone transfers; UDP for queries) [1, 2] |
| <b>67</b>  | UDP                | <b>DHCPS</b>      | DHCP Server port [2]                                                |
| <b>68</b>  | UDP                | <b>DHCPC</b>      | DHCP Client port [3]                                                |
| <b>69</b>  | UDP                | <b>TFTP</b>       | Trivial File Transfer Protocol [3]                                  |
| <b>80</b>  | TCP                | <b>HTTP</b>       | HyperText Transfer Protocol [1]                                     |
| <b>110</b> | TCP                | <b>POP3</b>       | Post Office Protocol (legacy mailbox access) [4]                    |
| <b>111</b> | TCP                | <b>RPCBIND</b>    | Maps RPC services to port numbers in UNIX-like environments [4]     |
| <b>123</b> | UDP                | <b>NTP</b>        | Network Time Protocol [3]                                           |
| <b>135</b> | TCP / UDP          | <b>MSRPC</b>      | Advertises RPC services in a Windows environment [3, 4]             |
| <b>137</b> | UDP                | <b>NETBIOS-NS</b> | NetBIOS Name Service (pre-                                          |

| Port       | Transport Protocol | Service Name        | Description                                                       |
|------------|--------------------|---------------------|-------------------------------------------------------------------|
|            |                    |                     | Windows 2000 file sharing) [3]                                    |
| <b>138</b> | UDP                | <b>NETBIOS-DGM</b>  | NetBIOS Datagram Service (pre-Windows 2000 file sharing) [3]      |
| <b>139</b> | TCP / UDP          | <b>NETBIOS-SSN</b>  | NetBIOS Session Service (pre-Windows 2000 file sharing) [3, 4]    |
| <b>143</b> | TCP                | <b>IMAP</b>         | Internet Mail Access Protocol [4]                                 |
| <b>161</b> | UDP                | <b>SNMP</b>         | Agent port for Simple Network Management Protocol [5]             |
| <b>162</b> | UDP                | <b>SNMP</b>         | Management station port for receiving SNMP trap messages [5]      |
| <b>443</b> | TCP                | <b>HTTPS</b>        | HyperText Transfer Protocol Secure [4]                            |
| <b>445</b> | TCP / UDP          | <b>MICROSOFT-DS</b> | Windows File Sharing (SMB over TCP/IP) on current networks [4, 5] |
| <b>500</b> | UDP                | <b>ISAKMP</b>       | Used to set up IPsec tunnels [6]                                  |
| <b>514</b> | UDP                | <b>SYSLOG</b>       | Server port for a syslog daemon [6]                               |
| <b>520</b> | UDP                | <b>RIP</b>          | Routing Information Protocol [6]                                  |
| <b>631</b> | UDP                | <b>IPP</b>          | Internet Printing Protocol [6]                                    |
| <b>993</b> | TCP                | <b>IMAPS</b>        |                                                                   |

| Port | Transport Protocol | Service Name | Description                                                          |
|------|--------------------|--------------|----------------------------------------------------------------------|
|      |                    |              | Internet Mail Access Protocol Secure [7]                             |
| 995  | TCP                | POP3S        | Post Office Protocol Secure [7]                                      |
| 1434 | UDP                | MS-SQL       | Microsoft SQL Server [6]                                             |
| 1723 | TCP                | PPTP         | Legacy VPN protocol with weak security implementation [7]            |
| 1900 | UDP                | UPNP         | Universal Plug and Play (autoconfiguration for smart appliances) [6] |
| 3306 | TCP                | MySQL        | MySQL database connection [2]                                        |
| 3389 | TCP                | RDP          | Remote Desktop Protocol [2]                                          |
| 4500 | UDP                | NAT-T-IKE    | IPsec traversal through a NAT gateway [8]                            |
| 5900 | TCP                | VNC          | Virtual Network Computing remote access service [2]                  |
| 8080 | TCP                | HTTP-PROXY   | HTTP Proxy Service or alternate port for HTTP [2]                    |

## General Port Classifications

The sources also note that the **Internet Assigned Numbers Authority (IANA)** categorizes ports into three ranges: \* **Well-known Ports:** 0 to 1023 [9] \* **Registered Ports:** 1024 to 49151 [9] \* **Dynamic (Private) Ports:** 49152 to 65535 [9]

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** "Name | Description | " - [2] **CompTIA+CySA++**



(CS0-003)+Study+Guide.docx.pdf: " | :— | :— | | **21** |" - [3] **CompTIA+CySA++**  
**(CS0-003)+Study+Guide.docx.pdf:** "P\*\* | File Transfer Protocol [1] | | **2**" - [4]  
 CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf: " | SSH/SFTP\*\* | Secure Shel"  
 - [5] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf, Source 2:** "SH [1] |  
 | **23** | TCP | **TELNET** | Unsecure remote admin" - [6] **Source 2:** "interface [1] |" - [7]  
**Source 3:** " TCP | **SMTP** " - [8] **Source 3:** "Mail Transfer Protocol [1] | |" - [9]  
**CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** " TCP / UDP | **DNS** |  
 Doma" - [10] **Source 4:** "stem (TCP for zone transfers; UDP for querie" - [11]  
**Source 4:** " **67** | UDP | **DHCP** | DHCP Server port [2] | | **68**" - [12] **Source 3:**  
 "**DHCP** | DHCP Clie" - [13] **Source 3, Source 4:** "**69** | **UDP** | **TFTP** | **Trivial File**  
**Transfer P**" - [14] **Source 3:** "**80** | **TCP** | **HTTP** | **HyperText Transfer Protoco**" -  
 [15] **Source 3:** " | **TCP** | **POP3** | **Post Office Protocol (legacy mailbox**" - [16]  
**Source 3, Source 4:** "\*" | **TCP** | **RPCBIND** | **Maps RPC services to port numbe**" -  
 [17] **Source 4:** "**-like environments** [4] | | " - [18] **Source 5:** "P | **NTP** | Network  
 Time Protocol [3] | | **135**" - [19] **Source 5:** " **UDP** | **MSRPC** | **Advertises RPC**  
**services in a Windows** " - [20] **Source 4:** " [3, 4] | | 137 | **UDP** | **NETB**" - [21]  
**Source 4, Source 5:** "Service (pre-Windows 2000 file sharing) [3] | | **138** | " - [22]  
**Source 6:** "**IOS-DGM** | **NetBIOS Datagram**" - [23] **Source 6:** "**e-Windows 2000**  
**file sharing)** [3]" - [24] **Source 6:** "**39** | TCP / UDP | **NETBIOS**" - [25] **Source 6:**  
 "**NetBIOS Session Service (p**" - [26] **Source 7:** "**2000 file sharing)** [3, 4] | |  
 143" - [27] **Source 7:** "\*|**IMAP** | **Internet Mail Acc**" - [28] **Source 6:** "[4] | | 161 |  
**UD**" - [29] **Source 7:** "\*" | **Agent port for Simple Network Management**  
**Protocol**" - [30] **Source 6:** "**162**\*\* | **UDP** | **SNMP** | **Management station port**  
**for receiving S**" - [31] **Source 2:** "**sages** [5] | | 443 | **T**" - [32] **Source 2:** "**PS**\*\* |  
**HyperText Transf**" - [33] **Source 8:** "**re** [4] | | 445 | **TCP / UDP** | **MI**" - [34]  
**Source 2:** " | **Windows File Sharing (SMB over TCP/IP) on**" - [35] **Source 2:**  
 "**[4, 5]** | | 500 | **UDP** | **ISAKMP** | **Used**" - [36] **Source 9:** "[6] | | \*" - [37] **Source**  
**9:** "**PP**\*\* | Intern" - [38] **Source 9:** " | | 993 |"

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**You (Thu Jul 2 13:25:21 2026)**

so there where few scan types of nmap ... just make me a table of those flags and a short description of them which include keyword which might be used to find correct answer in mcq question (strictly from source)

---

## NotebookLM (Thu Jul 2 13:25:21 2026)

Based on the provided study guide, here is a consolidated table of **Nmap flags**, their descriptions, and the specific keywords often used in multiple-choice questions to identify them.

### Nmap Discovery and Stealth Flags

| Flag                         | Description                | MCQ Keywords to Watch For                                                                                  |
|------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------|
| <b>-sn</b>                   | <b>Host Discovery Scan</b> | “Skip port scan”; identifies if hosts are up without scanning their ports. [1]                             |
| <b>-sL</b>                   | <b>List Scan</b>           | “ <b>Reverse-DNS query</b> ”; lists IP addresses and discovers associated host names. [1]                  |
| <b>-PS</b>                   | <b>TCP SYN Ping</b>        | Probes specific ports using a “ <b>TCP SYN packet</b> ” instead of ICMP. [1, 2]                            |
| <b>--scan-delay</b>          | <b>Sparse Scanning</b>     | Issues probes with “ <b>significant delays</b> ” to be stealthier and avoid IDS/IPS. [2]                   |
| <b>-Tn</b>                   | <b>Scan Timing</b>         | “ <b>0 is slowest</b> ” and “ <b>5 is fastest</b> ”; adjusts the speed of probes. [2]                      |
| <b>-sI</b>                   | <b>TCP Idle Scan</b>       | Stealth method that uses a “ <b>zombie</b> ” machine to hide the true identity of the scanner. [2, 3]      |
| <b>-f</b> (or <b>--mtu</b> ) | <b>Fragmentation</b>       | Splits the TCP header across multiple “ <b>IP datagrams</b> ” to make detection difficult for IDS/IPS. [3] |

## Nmap Port Scanning Flags

| Flag       | Description             | MCQ Keywords to Watch For                                                              |
|------------|-------------------------|----------------------------------------------------------------------------------------|
| <b>-sS</b> | <b>TCP SYN Scan</b>     | <b>“Half-open scan”</b> ; identifies port state without sending an ACK packet. [4]     |
| <b>-sT</b> | <b>TCP Connect Scan</b> | <b>“Three-way handshake scan”</b> ; completes the connection by sending an ACK. [4, 5] |
| <b>-sN</b> | <b>Null Scan</b>        | Sends a packet with the <b>“header bit set to zero”</b> . [5]                          |
| <b>-sF</b> | <b>FIN Scan</b>         | Sends an <b>“unexpected FIN packet”</b> . [5]                                          |
| <b>-sX</b> | <b>Xmas Scan</b>        | Sends a packet with <b>“FIN, PSH, and URG flags”</b> all set to one. [5]               |
| <b>-sU</b> | <b>UDP Scan</b>         | Sends a UDP packet and identifies state based on <b>“response or timeout”</b> . [5]    |
| <b>-p</b>  | <b>Port Range</b>       | Targets <b>“specified ports”</b> rather than the default top 1,000. [6]                |

## Nmap Fingerprinting and Output Flags

| Flag       | Description              | MCQ Keywords to Watch For                                                                  |
|------------|--------------------------|--------------------------------------------------------------------------------------------|
| <b>-sV</b> | <b>Service Detection</b> | Identifies “ <b>application name and version</b> ” and protocol. [7]                       |
| <b>-A</b>  | <b>Intensive Scan</b>    | An “ <b>aggressive</b> ” or intensive scan that includes OS detection and device type. [7] |
| <b>-oN</b> | <b>Normal Output</b>     | Saves results to a file in a standard text format. [8]                                     |
| <b>-oX</b> | <b>XML Output</b>        | Best for “ <b>integration with SIEM</b> ” products. [8]                                    |
| <b>-oG</b> | <b>Grepable Output</b>   | Saves results in a format that is easy to search with tools like <code>grep</code> . [8]   |

## Key Terminology for Nmap MCQ Success

- **CPE (Common Platform Enumeration):** A scheme for identifying hardware, OS, and applications. [7, 9]
- **NSE (Nmap Scripting Engine):** Uses “**Lua scripting**” to carry out detailed probes and vulnerability detection. [9]
- **Filtered State:** Indicates Nmap cannot probe the port, usually because a “**firewall is blocking**” the scan. [6, 10]

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** “or | :— | :— | :— | | **-sn** | **Host Discovery Scan** |”Skip ” - [2] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** ” are up without scanning their ports. [1] | | **-sL** |” - [3] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf, Source 2:** “nd discovers ass” - [4] **Source 2:** “pecific ports using a “**TCP SYN pa**” - [5] **Source 2\*\*:** “elay \*\* | \*\*Sparse Scanning\*\* ” - [6] **\*\*Source 2, Source 3\*\*:** “er and avoid IDS/IPS. [2] | | \*\* - [7] **\*\*Source 3\*\*:** “ of probes. [2] | | \*\* -sI \*\* | \*\*TCP Id” - [8] **\*\*Source 4\*\*:** “anner. [2, 3] | | \*\* -f \*\* (or -mtu ) | \*\*Fragmentat” - [9] **\*\*Source 4, Source 5\*\*:** “ **“IP datagrams”** to make detection difficu” - [10] **\*\*Source 5\*\*:** “l” - [11] **\*\*Source 5\*\*:** “r” - [12] **\*\*Source 5\*\*:** “ **“Half-open**

s" - [13] **Source 5**: "\*" - [14] **Source 6**: "dshake scan"; completes the conn" - [15] **Source 7**: " **FIN Scan**" - [16] **Source 7**: "cket". [5] | | **-sX** | **Xmas Scan** | Sends a packet w" - [17] **Source 8**: " and URG flags" all set to one. [5] | | **-sU`** - [18] **Source 8**: "tifies st" - [19] **Source 8**: "s" - [20] **Source 7, Source 9**: "Fingerprinting and Output Flags"

Flag | Description "

- [21] **Source 9**: " | | **-sV** | **Service Detection** | Identifies "ap"
- [22] **Source 6, Source 10**: "ssive" "

## You (Thu Jul 2 13:50:07 2026)

ok now i need you to help me like this there are several heading in the entire pdf whose short forms sometimes appear in the mcq questions answers all i want u to scan the entire pdf provide em the abbreviation like ueba it has a full form and give me a short description of them which keyword which if appears in the question might indicate there answer — please help me make the table also start by first 10 or 20 names and then when i say go u move to next set of 20 headings

## NotebookLM (Thu Jul 2 13:50:07 2026)

### Incorporating NIST Frameworks

I've just finished incorporating NIST-related definitions, including SP 800-53, AC, AA, IR, and RA, focusing on their significance as control families. I also added ISO 27001 as an international security standard and MISP as a threat intelligence sharing platform. The definitions are concisely capturing their key functions and implications.

## You (Thu Jul 2 13:54:31 2026)

ok listen there are several framework names in the pdf could u please make me a table of them and short description of there usage(mcq important keyword included if possible)

## **NotebookLM (Thu Jul 2 13:54:31 2026)**

Based on the sources provided, here is a table

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## **You (Thu Jul 2 13:54:51 2026)**

ok listen there are several framework names in the pdf could u please make me a table of them and short description of there usage(mcq important keyword included if possible) strictly from source uploaded

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## **NotebookLM (Thu Jul 2 13:54:51 2026)**

Based on the source material, here is a table of the various cybersecurity and operational frameworks, their usage, and the key terms frequently associated with them for exam purposes.

## Core Cybersecurity and Attack Frameworks

| Framework Name                            | Description / Usage                                                                                                                                         | MCQ Keywords                                                                                             |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>NIST SP 800-53</b>                     | A document providing security and privacy controls for <b>federal information systems</b> and organizations [1].                                            | "18 families of controls,"<br>"Technical, Operational, and Managerial classes" [1-3].                    |
| <b>ISO/IEC 27001</b>                      | An <b>international standard</b> and proprietary framework for specifying requirements for an information security management system [2, 4].                | "International standard,"<br>"Proprietary,"<br>"Requirements for ISMS" [2, 4].                           |
| <b>Lockheed Martin Kill Chain</b>         | Describes the <b>stages</b> by which a threat actor progresses a network intrusion [5].                                                                     | "7 steps: Recon, Weaponization, Delivery, Exploitation, Installation, C2, Actions on Objectives" [5, 6]. |
| <b>MITRE ATT&amp;CK</b>                   | A knowledge base listing and explaining specific <b>adversary tactics, techniques, and procedures</b> (TTPs) [7].                                           | "Adversary behaviors,"<br>"Knowledge base of tactics/ techniques" [7, 8].                                |
| <b>Diamond Model</b>                      | A framework for analyzing incidents by exploring relationships between four core features: <b>adversary, capability, infrastructure, and victim</b> [7, 9]. | "Relationships between core features," "Meta-features" [9].                                              |
| <b>NIST Cybersecurity Framework (CSF)</b> | A <b>risk-based framework</b> focused on IT security over IT service provision [10].                                                                        | "Identify, Protect, Detect, Respond, and Recover,"<br>"Core, Tiers, and Profiles" [10, 11].              |

## Compliance and Industry-Specific Frameworks

| Framework Name      | Description / Usage                                                                                                                           | MCQ Keywords                                                                |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>PCI DSS</b>      | Standards created by credit card companies to protect <b>sensitive payment card information</b> [12].                                         | “Major credit card companies,” “6 main categories/control objectives” [12]. |
| <b>CIS Controls</b> | A set of <b>best practice guidelines</b> and security controls organized into three categories: Basic, Foundational, and Organizational [13]. | “Top 20 Critical Security Controls,” “CIS Benchmarks” [13-15].              |
| <b>OWASP</b>        | Focuses on promoting and improving <b>web application security</b> [16].                                                                      | “Top Ten,” “Web application vulnerabilities” [16].                          |
| <b>ITIL</b>         | A framework for <b>IT service delivery</b> and management [17, 18].                                                                           | “Utility (fit for purpose),” “Service management” [18].                     |
| <b>COBIT</b>        | A framework used for <b>IT governance</b> and stipulated control selection [17, 19].                                                          | “Control selection and deployment,” “Regulatory compliance” [19].           |
| <b>OSS TMM</b>      | A framework for evaluating and improving the quality of <b>open source software</b> and its testing processes [20, 21].                       | “5 levels of testing maturity,” “Open source testing goals” [21, 22].       |



## Specialized Technology Frameworks

| Framework Name       | Description / Usage                                                                                                        | MCQ Keywords                                                                       |
|----------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>SCAP</b>          | A NIST framework outlining accepted practices for <b>automating vulnerability scanning</b> [23].                           | "Automating scanning," "Accepted practices," "SCAP-compliant" [23, 24].            |
| <b>Microsoft SDL</b> | A security framework for <b>application development</b> that supports dynamic development processes [25].                  | "Security Development Life Cycle," "Microsoft framework" [25].                     |
| <b>SASE</b>          | A network architecture that combines <b>network security and WAN capabilities</b> into a single cloud-based solution [26]. | "Software-defined networking (SDN)," "Secure Edge," "Network security + WAN" [26]. |
| <b>ESA</b>           | A framework for defining the <b>baseline, goals, and methods</b> used to secure a business [17].                           | "Enterprise Security Architecture," "Governance-based" [17].                       |

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**: "y and privacy con" - [2] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**, **Source 2**, **Source 3**: "rols for **federal information systems** and organizations [1]. | "18 famil" - [3] **Source 2**, **Source 4**: "l, and Managerial classes" [1-3]. | | **ISO/IEC 27001** | An **international standard** and proprie" - [4] **Source 2**, **Source 4**: "ary framework for specifying requirements for an information sec" - [5] **Source 5**: "standard," "Proprietary," "Requirements for ISMS" [2," - [6] **Source 5**, **Source 6**: "4]. | | **Lockheed Martin Kill Chain** | Describes the **stages** by which a threat actor progre" - [7] **Source 7**: "tions" - [8] **Source 7**, **Source 8**: "n Objectives" [5, 6]. | | **MITRE ATT&CK** | A knowledge base" - [9] **Source 7**, **Source 9** - [10] **Source 9**: "es" [7, 8]. | | **Diamond Model** | A framework for an" - [11] **Source 10**: "core features: **adversary, capability, infrastr**" - [12] **Source 10**, **Source 11**: "**cture, and victim** [7, 9]. | "Relationships between core features," "Meta-fea" - [13] **Source 12** - [14] **Source 12**: "over," "Core, Tiers, and Profiles" [10, 11]. |

## **\*\*Compliance and "**

| <b>[15] Source 13: "e</b> | <b>Description / Usage</b> | <b>MCQ Keywords</b> |
|---------------------------|----------------------------|---------------------|
| <b>PCI DSS</b>            | Standards crea"            |                     |

- [16] **Source 13, Source 14, Source 15:** "ed by credit card companies to protect \*\*sensitive pa"
- [17] **Source 16**
- [18] **Source 16:** " main categories/control objectives" [12]. |"
- [19] **Source 17, Source 18:** "ctice guidelin"
- [20] **Source 18:** "s\*\* and security controls organized into three ca"
- [21] **Source 17, Source 19:** "zational [13]. |"Top 20 Critica"
- [22] **Source 19:** " Security Controls," "CIS Benchmarks" [13-15]. | | \*\*OWASP\*\*"
- [23] **Source 20, Source 21:** "" "Web application vulner"
- [24] **Source 21, Source 22:** "bilities" [16]. | | **ITIL** | A framework for \*\*IT service"
- [25] **Source 23**
- [26] **Source 23, Source 24:** "ntrol selection [17, 19]. |"Control selection and deployment"
- [27] **Source 25:** "for evaluating and improving the quality of"
- [28] **Source 25:** "**open source software** and its testing processes [20,"
- [29] **Source 26:** "### \*\*Specialized Technology Frame"
- [30] **Source 26:** "orks\*\*"

Framework Name | Description / Usage | MCQ Keywords |  
:— | :-"

- [31] **Source 17:** "actices for \*\*automating "
- [32] **Source 17:** "ulnerability scanning\*\* [23]. |"Automating scanning,""

## **You (Thu Jul 2 14:00:18 2026)**

there are several short from of names used in the pdf which is used to make cysa plus mcqs questions examplen What regulation protects the privacy of student educational records? Your answer is incorrect SOX GLBA HIPAA Correct answer FERPA

now its hard for me to remeber sox full form glba full form

so could u make me another table for them write me the first 10 and then when i say move to the next 10

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### **NotebookLM (Thu Jul 2 14:00:18 2026)**

To help you master these for your exam, here is a table featuring the first 10 abbreviations and regulations found in your study notes, including the specific keywords that typically appear in CySA+ multiple-choice questions.

## **Key Abbreviations and Regulations (Set 1)**

| <b>Abbreviation</b> | <b>Full Name</b>                               | <b>Short Description / Usage</b>                                               | <b>MCQ Keywords</b>                                   |
|---------------------|------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>NIST</b>         | National Institute of Standards and Technology | A non-regulatory agency that provides security and privacy controls. [1]       | "Federal information systems," "families of controls" |
| <b>ISO</b>          | International Organization for Standardization | Develops and publishes international standards like ISO 27001. [2]             | "International standard," "Proprietary framework"     |
| <b>CIA</b>          | Confidentiality, Integrity, and Availability   | The core triad used to select and apply security controls. [3, 4]              | "Tenets of security," "Mitigate risk"                 |
| <b>OSINT</b>        | Open-Source Intelligence                       | Publicly available information gathered from websites and social media. [5, 6] | "Public records," "Aggregate and search"              |
| <b>ISAC</b>         | Information Sharing and Analysis Center        | Not-for-profit groups for sharing sector-specific threat intelligence. [5, 7]  | "Sector-specific," "Not-for-profit," "Best practices" |
| <b>APT</b>          | Advanced Persistent Threat                     | Sophisticated actors who maintain long-term covert access to a network. [8]    | "Persistence," "Maintain covert access"               |
| <b>PII</b>          | Personally Identifiable Information            | Data used to uniquely identify, contact, or impersonate an individual. [8, 9]  | "Identify or contact," "Data breach"                  |
| <b>STIX</b>         | Structured Threat Information eXpression       | A standard terminology used to describe Indicators of Compromise (IoCs). [10]  | "Standard terminology," "IoC relationships"           |
| <b>TAXII</b>        |                                                | A protocol used to automate the                                                | "Codified information,"                               |

| Abbreviation | Full Name                                           | Short Description / Usage                                                       | MCQ Keywords                                               |
|--------------|-----------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------|
|              | Trusted Automated eXchange of Indicator Information | exchange of threat intelligence data. [11]                                      | "Automate detection"                                       |
| <b>SIEM</b>  | Security Information and Event Management           | A system used for log management, analytics, and compliance reporting. [12, 13] | "Log review," "Security assurance," "Compliance reporting" |

Please let me know when you are ready to **go** to the next 10 abbreviations.

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** "— | | **NIST** | National Institute of Standards and Technology |" - [2] **Source 2:** "s," "families of controls" | | **ISO** | International Organiz" - [3] **Source 3, Source 4:** " "International standard," "Proprietary framework" | | **C** - [4] **Source 5, Source 6:** "ad used to select and apply security controls. [3, 4] | "Tenets of secu" - [5] **Source 5, Source 7:** "e information gathered from websites and social media. [5, 6] | "Publi" - [6] **Source 8:** " **Center** | **Not-for-profit groups for sharing sector-specific threat inte**" - [7] **Source 8, Source 9:** "APT | **Advanced Persistent Threat** | **Sophisticated actors who mainta**" - [8] **Source 10:** "ccess" | | PII\*\* | Personally Identifiable Information | Data used to " - [9] **Source 11:** "ach" | | **STIX** | Structured Threat Information eXpression | A stan" - [10] **Source 12, Source 13:** "nology," "IoC relationships" | | **TAXII** | Trusted Automated eXchang"

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**You (Thu Jul 2 14:01:11 2026)**

Go

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**NotebookLM (Thu Jul 2 14:01:11 2026)**

Here is the next set of 10 abbreviations and regulations found in your study notes, focusing on data governance, business continuity, and risk metrics.

## **Key Abbreviations and Regulations (Set 2)**

| Abbreviation   | Full Name                                           | Short Description / Usage                                                                                    | MCQ Keywords                                                               |
|----------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>GDPR</b>    | General Data Protection Regulation                  | A governance requirement that ensures the rights of subjects over their personal data. [1]                   | "72 hours breach notification," "informed consent," "withdraw consent" [2] |
| <b>SOX</b>     | Sarbanes-Oxley Act                                  | Sets requirements for the storage and retention of financial and business documents. [2]                     | "Financial and business operations," "document retention" [2]              |
| <b>FISMA</b>   | Federal Information Security Management Act         | Requires federal organizations in the US to adopt information assurance controls. [3]                        | "Federal organizations," "information assurance controls" [3]              |
| <b>HIPAA</b>   | Health Insurance Portability and Accountability Act | Protects the privacy of health information held by providers, hospitals, and insurers. [3]                   | "Health information," "healthcare providers," "insurance companies" [3]    |
| <b>GLBA</b>    | Gramm-Leach-Bliley Act                              | Requires financial institutions to provide customers with privacy notices and protect their information. [3] | "Financial institutions," "privacy notice," "customer information" [3]     |
| <b>PCI DSS</b> | Payment Card Industry Data Security Standard        | Standards focused on protecting credit card information and sensitive cardholder data. [4]                   | "Credit cards," "cardholder data," "quarterly scan" [4, 5]                 |
| <b>BIA</b>     | Business Impact Analysis                            | An activity used to identify organizational risks and their effect on                                        | "Mission critical operations," "system availability" [6]                   |



| Abbreviation | Full Name                  | Short Description / Usage                                                                         | MCQ Keywords                                                       |
|--------------|----------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|              |                            | mission-critical operations. [6]                                                                  |                                                                    |
| <b>MTD</b>   | Maximum Tolerable Downtime | The longest period of time a business can be inoperable before irrevocable failure occurs. [6, 7] | “Upper limit on recovery time,” “irrevocable business failure” [7] |
| <b>RTO</b>   | Recovery Time Objective    | The targeted length of time it takes to resume normal operations after an event. [8]              | “Length of time to resume,” “normal business operations” [8]       |
| <b>RPO</b>   | Recovery Point Objective   | The maximum amount of time that an organization can tolerate data being unrecoverable. [8]        | “Unrecoverable lost data,” “how long without your data” [8, 9]     |

Please let me know when you are ready to **go** to the next 10 abbreviations.

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** “- | : — | :— | | **GDPR** | General Data Protection Regulation | A governance requi” - [2] **Source 2:** “rement that ensures the rights of subjects over their personal data. [” - [3] **Source 2:** “ notification,” “informed consent,” “withdraw consent” [2] | | **SOX** | Sarbanes-Ox” - [4] **Source 2:** “ley Act | Sets requirements for the storage and retention” - [5] **Source 3:** “ancial and business operations,” “document retention” [2] | | **FISMA** | Federal” - [6] **Source 3:** “ Information Security Management Act | Requires federal o” - [7] **Source 3:** “ntrols. [3] |”Federal organizations,” “information assurance controls” [3] | | **HIPA**” - [8] **Source 3:** “**A\*\*** | **Health Insurance Portability and Accountability Act | Protect**” - [9] **Source 3:** “**formation held by providers, hospitals, and insurers. [3] |**”**Health information,**” “healthcare provid...” - [10] **Source 3:** “” “insurance companies” [3] | | **GLBA\*\*** | Gramm-Leach-Bliley Act ” - [11] **Source 4:** “ers with privacy notices and protect their information. [3] |”Financial institutions,” - [12] **Source 4, Source 5:** “” “privacy notice,” “customer information” [3] | | ” - [13] **Source 6:** “Industry Data Security Standard | Standards focused on protecting credit card information and sens” - [14] **Source 6:** “itive cardholder data. [4] |”Credit cards,” “cardho” - [15] **Source 6, Source 7:** “[4, 5] | | **BIA** | Business Impact Analysis | An activity used to identify organizationa” - [16] **Source 7:** “l risks and

their effect on mission-critical operations. [6] | " - [17] **Source 8:** "ons," "system availability" [6] | | **MTD** | Maximum Tolerable Downtime | The l" - [18] **Source 8:** "ongest period of time a business can be inoperable befor" - [19] **Source 8:** "s. [6, 7] | "Upper limit on recovery time," "irrevocable business failure" [7] | | **R" - [20]** Source 8, Source 9\*\*: "TO\*\* | Recovery Time Objective | The targeted length of"

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**You (Thu Jul 2 15:41:03 2026)**

just before the next state tell me how many more may i expect

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**NotebookLM (Thu Jul 2 15:41:03 2026)**

### **Estimating Table Quantity**

I'm now focused on anticipating the remaining tables in the sequence. Reviewing our conversation, I'm analyzing the provided source to gauge how many more abbreviations are likely to be detailed. My analysis aims to accurately predict the total number of items left to process, giving a helpful estimate.

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**You (Thu Jul 2 15:41:22 2026)**

just before the next set tell me how many more may i expect

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**NotebookLM (Thu Jul 2 15:41:22 2026)**

Based on the source material, you can expect approximately **4 more sets** of abbreviations (around 40 more terms) to cover the key technical, forensic, and risk-related acronyms used in the CySA+ objectives.

Here is the next set, focusing on operational teams, attack indicators, and vulnerability standards:

## **Key Abbreviations and Regulations (Set 3)**

| <b>Abbreviation</b>    | <b>Full Name</b>                         | <b>Short Description / Usage</b>                                                                   | <b>MCQ Keywords</b>                                                                  |
|------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>SOC</b>             | Security Operations Center               | A location where professionals monitor and protect critical information assets [1].                | "Monitor and protect assets," "separation of signal from noise" [1, 2]               |
| <b>CSIRT</b>           | Computer Security Incident Response Team | A specialized team that provides incident response services to an organization [3].                | "Performing incident response," "responding to a breach" [2, 3]                      |
| <b>IoC</b>             | Indicator of Compromise                  | Evidence or signs that an attack on a network or asset was successful [4, 5].                      | "Evidence an attack was successful," "sign an asset has been attacked" [4, 5]        |
| <b>IoA</b>             | Indicator of Attack                      | Evidence that a network intrusion attempt is currently in progress [4].                            | "Intrusion attempt in progress" [4]                                                  |
| <b>C2 (or C&amp;C)</b> | Command and Control                      | The infrastructure used by an attacker to issue instructions to compromised "zombie" hosts [6, 7]. | "Beaconing," "issue commands to zombies," "DGA or fast flux" [6, 7]                  |
| <b>TTP</b>             | Tactics, Techniques, and Procedures      | A knowledge base listing and explaining specific adversary behaviors [8].                          | "Adversary behaviors," "MITRE ATT&CK" [8]                                            |
| <b>CVSS</b>            | Common Vulnerability Scoring System      | A standard used to rank the severity of vulnerabilities to prioritize response [9].                | "Prioritizing response actions," "base, temporal, and environmental metrics" [9, 10] |
| <b>SCAP</b>            | Security Content Automation Protocol     | A NIST framework of accepted practices                                                             |                                                                                      |

| Abbreviation | Full Name                                             | Short Description / Usage                                                              | MCQ Keywords                                                    |
|--------------|-------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------|
|              |                                                       | for automating vulnerability scanning [11].                                            | "Automating scanning," "adhering to standards" [11]             |
| <b>OVAL</b>  | Open Vulnerability and Assessment Language            | An XML schema used to describe a system's security state and query reports [12].       | "XML schema," "describing system security state" [12]           |
| <b>XCCDF</b> | Extensible Configuration Checklist Description Format | An XML schema for developing and auditing best-practice configuration checklists [12]. | "Best-practice configuration checklists," "auditing rules" [13] |

Please let me know when you are ready to **go** to the next set.

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** " | | : — | :— | :— | :— | | **SOC** | Security Operations Center | A lo" - [2] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf,** **Source 2:** "ation where professionals monitor and protect critical informat" - [3] **Source 3:** "," "separation of signal from noise" [1, 2] | | **CSIRT** | Computer Security " - [4] **Source 2, Source 3:** "ncident Response Team | A specialized team that provides" - [5] **Source 4, Source 5:** "s to an organization [3]. | "Performing incident response," "respon" - [6] **Source 4, Source 5:** "g to a breach" [2, 3] | | **IoC** | Indicator of Compromise | Evidence" - [7] **Source 4:** "ck on a network or asset was successful [4, 5]. | "Evidence an att" - [8] **Source 4:** "ck was successful," "sign an as" - [9] **Source 6, Source 7:** " | | **IoA** | Indicator of Attack | Evidence that a network intrusion attempt is currently" - [10] **Source 6, Source 7:** "in progress [4]. | "Intrusion attempt in progress" [4] | | "\*" - [11] **Source 8:** " The infrastructure used by an attacker to issue instructions to com" - [12] **Source 8:** "romised"zombie" hosts [6, 7]. | "Bea" - [13] **Source 9:** "DGA or fast flux" [6, 7] | | **TTP** | Tactics, Techniques, and Procedures | A" - [14] **Source 9, Source 10:** "knowledge base listing and explaining specific adversary behaviors [8]. | "A" - [15] **Source 11:** " | | **CVSS** | Common Vulnerability Scoring System | A standard used to rank" - [16] **Source 11:** "the severity of vulnerabilities to prioritize" - [17] **Source 12:** "," "base, temporal, and environmental metrics" [9, 10] | | **SCAP** | Secu" - [18] **Source 12:** "ity Content Automation Protocol | A NIST framewo" - [19] **Source 12:** "ning [11]. | "Automating scanning," "adhering to

standards" [11] | | **OVAL** | " - [20] **Source 13**: "pen Vulnerability and Assessment Language | An XML schema"

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**You (Thu Jul 2 15:47:22 2026)**

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**NotebookLM (Thu Jul 2 15:47:22 2026)**

Here is the next set of abbreviations from the study notes, focusing on network security systems and formal organizational agreements.

## **Key Abbreviations and Regulations (Set 4)**

| Abbreviation | Full Name                   | Short Description / Usage                                                                                           | MCQ Keywords                                               |
|--------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>IDS</b>   | Intrusion Detection System  | A system that scans, audits, and monitors security infrastructure for signs of attacks in progress [1].             | "Monitors for signs of attacks," "scans and audits"        |
| <b>IPS</b>   | Intrusion Prevention System | A system that monitors for attacks and can <b>actively block</b> them from succeeding [2].                          | "Actively block attacks," "prevent access"                 |
| <b>WAF</b>   | Web Application Firewall    | A system whose logs can be analyzed for malicious code in HTTP request headers and URLs [3].                        | "HTTP request headers," "malicious URL parameters"         |
| <b>NAC</b>   | Network Access Control      | Protocols and policies used to authenticate users and evaluate device integrity before a connection is allowed [4]. | "Device integrity," "authenticate users before connection" |
| <b>BYOD</b>  | Bring Your Own Device       | A policy allowing employees to use personal smartphones and tablets for work and network connectivity [5].          | "Personal smartphones," "corporate network connection"     |
| <b>SLA</b>   | Service Level Agreement     | A <b>contractual agreement</b> specifying the detailed terms under which a service is provided [6, 7].              | "Contractual agreement," "detailed terms of service"       |
| <b>MOU</b>   |                             |                                                                                                                     |                                                            |



| Abbreviation | Full Name                          | Short Description / Usage                                                                                      | MCQ Keywords                                                   |
|--------------|------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
|              | Memorandum of Understanding        | A preliminary agreement expressing an intent to work together that is <b>not legally binding</b> [6].          | "Intent to work together," "not legally binding"               |
| <b>ISA</b>   | Interconnection Security Agreement | An agreement used (often by federal agencies) to commit to implementing security controls with a supplier [8]. | "Commit to implementing security controls," "federal agencies" |
| <b>NDA</b>   | Non-Disclosure Agreement           | A legal agreement used to protect confidential information shared between parties [8].                         | "Protect confidential information," "legal agreement"          |
| <b>AUP</b>   | Acceptable Use Policy              | A policy that governs how employees use company equipment and internet services [9].                           | "Use of company equipment," "employee behavior"                |

Please let me know when you are ready to **go** to the next set.

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** "— | : — | :— | :— | | **IDS** | Intrusion Detection System | A system that scans, audits, a" - [2] **Source 2:** "| Intrusion Preventi" - [3] **Source 3:** "k\*\* them from succeeding [2]. |"Actively block attacks," "prevent access" | | **WAF** " - [4] **Source 4:** "us code in HTTP request headers and URLs [3]. |"HTTP request headers," "malicious URL parameters" |..." - [5] **Source 5:** "aluate device integrity before a connection is allowed [4]. |"Device integrity," "authenticate user..." - [6] **Source 6, Source 7:** "artphones and tablets for work and network connectivity [5]. |" - [7] **Source 6** - [8] **Source 8:** "U\*\* | Memorandum of Understanding | A preliminary agreement expressing an intent to work together th..." - [9] **Source 8:** "\*\* | Interconnection Security Agreement | An agreement used (often by federal age" - [10] **Source 9:** "mit to implementing security controls," "federal agencies" | | **NDA** | Non-Di"

**You (Thu Jul 2 15:50:32 2026)**

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**NotebookLM (Thu Jul 2 15:50:32 2026)**

Here is the next set of 10 abbreviations from the source material, focusing on data privacy, cloud services, and automation.

## **Key Abbreviations and Regulations (Set 5)**

| Abbreviation | Full Name                                        | Short Description / Usage                                                                       | MCQ Keywords                                           |
|--------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>SPI</b>   | Sensitive Personal Information                   | Information about a subject's opinions or beliefs afforded special legal protection [1, 2].     | "Opinions or beliefs,"<br>"Specially protected status" |
| <b>PHI</b>   | Personal Health Information                      | Information that identifies someone as the subject of medical or insurance records [2].         | "Medical records,"<br>"Hospital results"               |
| <b>SOAR</b>  | Security Orchestration, Automation, and Response | A system primarily used for incident response and automating data enrichment [3].               | "Data enrichment,"<br>"Automating incident response"   |
| <b>SAML</b>  | Security Assertions Markup Language              | An XML-based format used to exchange authentication data for SSO and federations [4].           | "XML-based SSO,"<br>"Federated identity management"    |
| <b>REST</b>  | Representational State Transfer                  | A loose architectural style for creating web services that is often contrasted with SOAP [5].   | "RESTful APIs,"<br>"Loose constraints"                 |
| <b>JWT</b>   | JSON Web Token                                   | A token format containing a header, payload, and signature used to pass information safely [6]. | "Token format,"<br>"Header, payload, and signature"    |
| <b>IaC</b>   | Infrastructure as Code                           | A provisioning architecture where resources are deployed via scripted automation                | "Scripted provisioning,"<br>"Orchestration runbooks"   |

| Abbreviation | Full Name                                    | Short Description / Usage                                                                   | MCQ Keywords                                                |
|--------------|----------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------|
|              |                                              | and orchestration [7].                                                                      |                                                             |
| <b>CI/CD</b> | Continuous Integration / Continuous Delivery | Methods where code updates are tested and committed to production rapidly and often [8, 9]. | "Rapid commits," "Tested multiple times per day"            |
| <b>SaaS</b>  | Software as a Service                        | A cloud model providing a complete application service, including the hardware and OS [10]. | "Complete application service," "integrity of the platform" |
| <b>VPC</b>   | Virtual Private Cloud                        | A private, isolated network instance provisioned within a public cloud environment [11].    | "Private network in public cloud," "Isolate resources"      |

Please let me know when you are ready to **go** to the next set.

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**,  
**Source 2:** " | :- | :- | :- | | **SPI** | Sensitive Personal Information | Information abo" -  
[2] **Source 2:** "inions or beliefs," "Specially protected status" | | **PHI** | Personal Health Inf" - [3] **Source 3:** "rds [2]. | "Medical records," "Hospital results" | | **SOAR** | Security Orc" - [4] **Source 4:** "automating data enrichment [3]. | "Data enrichment," "Automating incident respon" - [5] **Source 5:** " exchange authentication data for SSO and federations [4]. | "XML-based SSO," "Federated" - [6] **Source 6:** " State Transfer | A loose architectural style for creating web services that is often cont" - [7] **Source 7:** "JSON Web Token | A token format containing a header, payload, and signature used to pass informati" - [8] **Source 8, Source 9:** "ode | A provisioning architecture where resources are deployed via scripted automat" - [9] **Source 10:** "oks" | | **CI/CD** | Continuous Integration / Continuous Delivery | Methods where cod" - [10] **Source 11:** "commits," "Tested multiple times per day" | | **SaaS** | Software as a Service | A"

**You (Thu Jul 2 15:51:53 2026)**

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**NotebookLM (Thu Jul 2 15:51:53 2026)**

Here is the next set of 10 abbreviations from the source material, focusing on endpoint forensics, email security, and vulnerability identification.

## **Key Abbreviations and Regulations (Set 6)**

| Abbreviation | Full Name                  | Short Description / Usage                                                                      | MCQ Keywords                                                      |
|--------------|----------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>PE</b>    | Portable Executable        | The file format for Windows executables, DLLs, and drivers. [1]                                | "starts with 4D 5A (HEX)," "MZ in ASCII," "TV in Base64" [1]      |
| <b>PID</b>   | Process Identification     | A unique identification number assigned to every process launched in a Linux system. [2]       | "unique identification number," "systemd always has PID 1" [2]    |
| <b>PPID</b>  | Parent Process ID          | The unique identification number of the process that launched another process. [3]             | "parent/child relationship," "pstree command" [3]                 |
| <b>TGT</b>   | Ticket-Granting Ticket     | A Kerberos credential used by users to access services within an Active Directory domain. [4]  | "generated by krbtgt hash," "access services within Kerberos" [4] |
| <b>KDC</b>   | Key Distribution Center    | A network service that issues TGTs and service tickets in a Kerberos environment. [4]          | "Kerberos architecture," "Authentication Server" [4]              |
| <b>SPF</b>   | Sender Policy Framework    | A DNS record that identifies which mail servers are authorized to send email for a domain. [5] | "only one allowed per domain," "identify authorized hosts" [5]    |
| <b>DKIM</b>  | DomainKeys Identified Mail | A cryptographic authentication mechanism that uses a public key                                | "cryptographic authentication," "public key in DNS record" [5]    |



| Abbreviation  | Full Name                                                       | Short Description / Usage                                                                       | MCQ Keywords                                                             |
|---------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|               |                                                                 | published in DNS to verify mail. [5]                                                            |                                                                          |
| <b>DMARC</b>  | Domain-Based Message Authentication, Reporting, and Conformance | A framework used to ensure that SPF and DKIM are properly applied to outgoing mail. [5]         | "ensuring proper application of SPF and DKIM" [5]                        |
| <b>S/MIME</b> | Secure/Multipurpose Internet Mail Extensions                    | A protocol for sending digitally signed and encrypted messages using a digital certificate. [6] | "digital certificate," "Jason sends Mary his digital certificate" [6, 7] |
| <b>CVE</b>    | Common Vulnerabilities and Exposures                            | A standard scheme for identifying known software vulnerabilities. [8]                           | "CVE-YYYY-####," "format for identifying vulnerabilities" [8]            |

Please let me know when you are ready to **go** to the next set.

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** "ds | | :— | :— | :— | :— | | **PE** | Portable Exec" - [2] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** "utable | The file format for Windows executables, DLLs," - [3] **Source 2:** "s with 4D 5A (HEX)," "MZ in ASCII," "TV in Base64" [1] | | **PID** | Process Identif" - [4] **Source 2:** "ication | A unique identification number assigned to every" - [5] **Source 3:** "a Linux system. [2] | "unique identification number," "systemd always has PID" - [6] **Source 3:** "1" [2] | | **PPID** | Parent Process ID | The" - [7] **Source 4:** "mber of the process that launched another process. [3] | "parent/child relationship," "ps" - [8] **Source 4:** "tree command" [3] | | **TGT** | Ticket-Granting Ticket | A Ke" - [9] **Source 4:** "users to access services within an Active Directory domain. [4] | "generated by k" - [10] **Source 4:** "rbtgt hash," "access services within Kerberos" [" - [11] **Source 5:** "ibution Center | A network service that issues TGTs and service tickets in a Kerberos envi" - [12] **Source 5:** "ronment. [4] | "Kerberos architecture," "Authentication Se" - [13] **Source 5:** " Policy Framework | A DNS record that identifies which mail servers are authorized to send email" - [14] **Source 5:** " for a domain. [5] | "only one allowed per domain," "ident" - [15] **Source 5:** "| A cryptographic authentication mechanism that uses a public key published in DNS" - [16] **Source 5:** "to verify mail. [5] | "cryptographic authenti" - [17] **Source 6:** "C\*\* | Domain-Based Message

Authentication, Reporting, and Conformance | A framework used to" - [18] **Source 6, Source 7:** " ensure that SPF and DKIM are properly applied to outgoing mail. " - [19] **Source 8:** "PF and DKIM" [5] | | **S/MIME** | Secure/Multipurpose Internet Ma" - [20] **Source 8:** "il Extensions | A protocol for sending digitally signed a"). | "identifying hardware devices," "MITRE scheme" | | **NSE** | Nmap Scripting Engine" - [3] **Source 3:** "etection [2]. | "Lua scripting," "detailed probes," "OS detection" | | **NVD** | National Vulnerab" - [4] **Source 4:** "tions [3]. | "NIST superset of CVE," "criticality metrics (CVSS)" | | **CAPEC** | Common Attack" - [5] **Source 5, Source 6:** "s specific attack patterns focused on application security [4]. | "classifies attack" - [6] **Source 7, Source 8:** " Beta testing performed by end users to prove a program is usable and fit-for-" - [7] **Source 9:** "DA\*\* | Interactive Disassembler | A popular cross-platform disassembler and" - [8] **Source 10:** "sassembler" | | **ZAP** | Zed Attack Proxy | An OWASP open-source interception proxy and w" - [9] **Source 10:** ") mode" | | **PaaS** | Platform as a Service | A cloud model providing the hardware a" - [10] **Source 11, Source 12:** "aS and IaaS," "hardware and software for a service" | | **IaaS** | Infrastructure as a Servi"

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**You (Thu Jul 2 15:54:02 2026)**

how many more to go

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**NotebookLM (Thu Jul 2 15:54:02 2026)**

Based on the source material, there are approximately **3 more sets** (around 30 terms) remaining to comprehensively cover the important abbreviations, technical acronyms, and forensic terms mentioned in your study notes.

Here is the next set, focusing on network forensics, cloud-specific protocols, and attack variations:

## **Key Abbreviations and Regulations (Set 8)**

| Abbreviation | Full Name                          | Short Description / Usage                                                                        | MCQ Keywords                                              |
|--------------|------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>SPAN</b>  | Switched Port Analyzer             | A feature on a switch that copies traffic from one or more ports to another for analysis.        | "mirroring traffic," "copying ingress/egress"             |
| <b>FPC</b>   | Full Packet Capture                | Capturing the entire packet, including header and payload, for all network traffic.              | "entire packet," "complete record of events"              |
| <b>JSON</b>  | JavaScript Object Notation         | A standard text-based format for representing structured data, used by tools like Zeek and APIs. | "text-based data format," "RESTful messages"              |
| <b>SNMP</b>  | Simple Network Management Protocol | A protocol used to poll and manage network appliances like routers and switches.                 | "polling appliances," "community strings," "port 161/162" |
| <b>DRDoS</b> | Distributed Reflection DoS         | An attack that uses an amplification factor to increase bandwidth sent to a victim.              | "amplification factor," "spoofs victim IP"                |
| <b>CASB</b>  | Cloud Access Security Broker       | An enterprise management software that sits between consumers and cloud service providers.       | "security policy enforcement," "forward or reverse proxy" |
| <b>IAM</b>   | Identity and Access Management     | Policies and technologies to ensure the right users have appropriate access to resources.        | "managing user permissions," "cloud buckets"              |

| Abbreviation | Full Name                       | Short Description / Usage                                                                       | MCQ Keywords                                             |
|--------------|---------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>PAM</b>   | Privileged Access Management    | A solution used to mitigate risk by restricting the time and use of administrative credentials. | "mitigate insider threat," "time-restricted credentials" |
| <b>DGA</b>   | Domain Generation Algorithm     | An algorithm used by malware to create a large list of domain names to use as C2 points.        | "random domain names," "beaconing infrastructure"        |
| <b>EDR</b>   | Endpoint Detection and Response | A system that monitors and records activity on endpoints to detect and respond to threats.      | "tuning false positives," "monitoring end-user devices"  |

Please let me know when you are ready to **go** to the next set.

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**NotebookLM (Thu Jul 2 15:54:40 2026)**

Here is the next set of 10 abbreviations from the source material, focusing on application vulnerabilities, name resolution protocols, and business continuity. There are approximately **2 more sets** remaining to complete the collection.

## **Key Abbreviations and Regulations (Set 9)**

| Abbreviation | Full Name                            | Short Description / Usage                                                                                       | MCQ Keywords                                                                                  |
|--------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>XSS</b>   | Cross-Site Scripting                 | A malicious script injected into a trusted website designed to compromise browsing clients.                     | "script in link," "trusted site security," "circumventing browser model" [1]                  |
| <b>SQLi</b>  | SQL Injection                        | An attack where malicious SQL code is inserted into form fields or POST data to manipulate a database.          | "' OR 1=1;" "database reads," "least privilege for database" [2, 3]                           |
| <b>CSRF</b>  | Cross-Site Request Forgery           | An attack that tricks a user's browser into performing unwanted actions on a site where they are authenticated. | "OAuth2 vulnerability," "unauthorized actions," "open redirects" [4]                          |
| <b>SSRF</b>  | Server-Side Request Forgery          | An attack where a web application is tricked into making unauthorized requests to internal or external systems. | "request on behalf of web app," "access internal databases," "retrieve sensitive data" [5, 6] |
| <b>XXE</b>   | XML External Entity                  | An attack that embeds a request for a local resource within an XML document to exploit weak parsers.            | "XML encodes entities," "request for a local resource" [7, 8]                                 |
| <b>LLMNR</b> | Link-Local Multicast Name Resolution | A protocol for name resolution used on local links when DNS is unavailable; often targeted for poisoning.       | "poisoning responses," "name resolution," "Responder tool" [9]                                |
| <b>MDNS</b>  | Multicast DNS                        | A protocol used to resolve hostnames                                                                            |                                                                                               |

| Abbreviation | Full Name                          | Short Description / Usage                                                                                     | MCQ Keywords                                                                   |
|--------------|------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
|              |                                    | to IP addresses within small networks without a name server.                                                  | "name resolution requests," "Responder tool" [9]                               |
| <b>DLP</b>   | Data Loss Prevention               | Policies and technologies designed to ensure sensitive information is not sent outside the corporate network. | "prevent insider threat," "authorized access," "enforcement technologies" [10] |
| <b>EAP</b>   | Extensible Authentication Protocol | A framework for encapsulating authentication communications, frequently used with 802.1X.                     | "802.1X standard," "encapsulating communications," "EAPOL" [11]                |
| <b>MEF</b>   | Mission Essential Function         | A business process or organizational activity so critical it cannot be deferred for more than a few hours.    | "too critical to defer," "business processes," "system assessment" [12]        |

Please let me know when you are ready to **go** to the next set.

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** "injected into a trusted website designed to compromise browsing clients." - [2] **Source 2, Source 3:** "k where malicious SQL code is inserted into form fields or PO" - [3] **Source 4:** "rgery | An attack that tricks a user's browser into performing u" - [4] **Source 5, Source 6:** "Server-Side Request Forgery | An attack where a web application is tricked into making" - [5] **Source 7, Source 8:** "trieve sensitive data" [5, 6] | | **XXE** | XML Extern" - [6] **Source 9:** "t for a local resource" [7, 8] | | **LLMNR** | Link-Local " - [7] **Source 9:** "eted for poisoning. |"poisoning responses,"" - [8] **Source 10:** "small networks without a name server. |"name resolution requests," "Resp" - [9] **Source 11:** " sent outside the corporate network. | "prevent insider th" - [10] **Source 12:** "ating authentication communications, frequently used with 802.1X."



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**NotebookLM (Thu Jul 2 15:55:53 2026)**

Here is the next set of 10 abbreviations from your study notes, focusing on system defense, industrial control systems, and organizational management. This is the **penultimate set**, meaning there is only one more set remaining after this one.

## **Key Abbreviations and Regulations (Set 10)**

| Abbreviation       | Full Name                                 | Short Description / Usage                                                                                                | MCQ Keywords                                                                   |
|--------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>CISO</b>        | Chief Information Security Officer        | A senior-level executive who oversees an organization's information and technology security [1].                         | "Senior-level executive," "oversees technology security"                       |
| <b>BCP</b>         | Business Continuity Plan                  | A plan that outlines how an organization will continue to operate during and after a disaster [2, 3].                    | "Resuming operations," "NIST SP 800-34," "contingency planning"                |
| <b>SOP</b>         | Standard Operating Procedure              | A set of step-by-step instructions compiled by an organization to help workers carry out complex routine operations [4]. | "Playbook," "step-by-step," "routine operations"                               |
| <b>HIDS / NIDS</b> | Host / Network Intrusion Detection System | Systems that monitor for malicious activity at the individual host or network level [5].                                 | "Host-based," "Network-based," "Lockheed Martin Kill Chain mapping"            |
| <b>ASLR</b>        | Address Space Layout Randomization        | A memory-protection process that randomly arranges the address space positions of key data areas [6].                    | "Memory-protection," "protect against buffer overflows," "randomize addresses" |
| <b>DEP</b>         | Data Execution Prevention                 | A security feature that prevents code from being executed from non-executable memory regions [5, 6].                     | "Non-executable memory," "prevent execution," "buffer overflow protection"     |
| <b>ICS</b>         |                                           |                                                                                                                          |                                                                                |

| Abbreviation | Full Name                                | Short Description / Usage                                                                                               | MCQ Keywords                                                            |
|--------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
|              | Industrial Control System                | A collective term for types of control systems used for industrial process control, like manufacturing [7].             | "Industrial processes," "manufacturing," "embedded devices"             |
| <b>SCADA</b> | Supervisory Control and Data Acquisition | A type of industrial control system that manages large-scale, multiple-site equipment over geographic regions [7].      | "Large-scale management," "geographic regions," "manage plant devices"  |
| <b>PLC</b>   | Programmable Logic Controller            | An industrial computer used for the automation of typically industrial electromechanical processes [7].                 | "Industrial computer," "manage equipment," "embedded in SCADA"          |
| <b>RTOS</b>  | Real-Time Operating System               | An operating system that prioritizes deterministic execution to ensure consistent response for time-critical tasks [8]. | "Time-critical tasks," "deterministic execution," "consistent response" |

Please let me know when you are ready to **go** to the final set.

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf:** " :— | :— | :— | | **CISO** | Chief Information Security Officer | A senior-level exec" - [2] **Source 2, Source 3:** "ior-level executive," "oversees technology security" | | **BCP** | Business Continuity Plan |" - [3] **Source 4:** "[2, 3]. |"Resuming operations," "NIST SP 800-34," "contingency planning" | | **SOP** | Standard Ope..." - [4] **Source 5:** "routine operations [4]. |"Playbook," "step-by-step," "routine operations" | | **HI**" - [5] **Source 6:** "he individual host or network level [5]. |"Host-based," "Network-based," "Lockheed Martin Kill" - [6] **Source 5, Source 6:** "y arranges the address space positions of key data areas [6]. |"Memory-protection," "protec" - [7] **Source 7:** " feature that prevents code from being executed from non-executable memory regions [5, 6]. | "Non-ex..." - [8] **Source 7:** " |

**A collective term for types of control systems used for industrial process control, like manufact...” - [9] Source 7: ” Data Acquisition | A type of industrial control system that manages large-scale, multiple-site eq” - [10] Source 8: “e plant devices” | | PLC\*\* | Programmable Logic Controller | An industrial computer used for the a...”**

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**You (Thu Jul 2 15:56:36 2026)**

go

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**NotebookLM (Thu Jul 2 15:56:36 2026)**

Here is the **final set** of abbreviations from the source material, focusing on industrial technology, change management, and advanced network architectures.

## **Key Abbreviations and Regulations (Set 11)**

| <b>Abbreviation</b> | <b>Full Name</b>                      | <b>Short Description / Usage</b>                                                     | <b>MCQ Keywords</b>                                         |
|---------------------|---------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>OT</b>           | Operational Technology                | A communications network designed to implement an industrial control system.         | “industrial control,” “embedded system threats” [1, 2]      |
| <b>CAN</b>          | Controller Area Network               | A digital serial communications network used to connect subsystems within vehicles.  | “vehicle subsystems,” “serial data communications” [3]      |
| <b>OBD-II</b>       | Onboard Diagnostics                   | The primary external interface for vehicle networks, often targetable for exploits.  | “external interface,” “vehicular exploit vector” [3]        |
| <b>COSO</b>         | Committee of Sponsoring Organizations | A framework providing guidance on various governance-related topics.                 | “governance-related guidance,” “internal control” [4]       |
| <b>V&amp;V</b>      | Verification and Validation           | A process ensuring a security system meets design goals and is fit for purpose.      | “compliance-testing,” “fit for purpose” [5, 6]              |
| <b>CSI</b>          | Continual Service Improvement         | A process used to identify and implement changes to improve organizational services. | “improve services,” “identify and implement changes” [7, 8] |
| <b>CAB</b>          | Change Advisory Board                 | A body responsible for reviewing and approving major or significant changes.         | “approve major changes,” “significant changes” [9]          |
| <b>RFC</b>          | Request for Change                    | A formal document listing the reason for                                             |                                                             |

| Abbreviation | Full Name                   | Short Description / Usage                                                         | MCQ Keywords                                                              |
|--------------|-----------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|              |                             | a change and its implementation procedures.                                       | "reason for change," "implementation procedures" [9]                      |
| <b>OIDC</b>  | OpenID Connect              | An authentication protocol implemented as special OAuth flows to verify identity. | "authentication protocol," "verify identity," "OAuth flow" [10, 11]       |
| <b>SDN</b>   | Software Defined Networking | A network architecture that separates the control plane from the data plane.      | "separate control plane," "data plane," "virtualized networking" [12, 13] |

This concludes the sets of abbreviations and key terms from your study notes. Each of these 110 terms is critical for recognizing correct answers and identifying the context of scenarios in the **CySA+ (CS0-003)** exam. Good luck with your final review!

**References:** - [1] **CompTIA+CySA++(CS0-003)+Study+Guide.docx.pdf**, **Source 2:** "communications network designed to implement an i" - [2] **Source 3:** "rea Network | A digital serial communications netw" - [3] **Source 3:** " **OBD-II** | Onboard Diagnostics | The primary " - [4] **Source 4:** "ploit vector" [3] | | **COSO** | Committee of Spo" - [5] **Source 5, Source 6:** "lated guidance," "internal control" [4]" - [6] **Source 7, Source 8:** "purpose. |"compliance-testing," "fit for purpose" [" - [7] **Source 9:** " to improve organizational services. | "improv" - [8] **Source 9:** "ible for reviewing and approving major or signif" - [9] **Source 10, Source 11:** "e | A formal document listing the reason for a change and" - [10] **Source 12, Source 13:** "D Connect | An authentication protocol implemented as special OA"].

## 4. Annual Loss Expectancy (ALE)

- **Formula: ALE = SLE x ARO** [3]
- **Terms:..."**
- [13] **Source 4**